

Compilers and Simulators

For Programming of Microcontrollers

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An embedded system consists of various components like RAM, flash memory, regulators, various ports and JTAG. But the most important device within an embedded system is the microcontroller.

A microcontroller is a system-on-a-chip that performs a particular task. It consists of a CPU, RAM, ROM, I/O ports and timers. Unlike a microprocessor that is designed for general-purpose computing, for example, our PC or laptops, a microcontroller does a specific task.

According to Shola Aiyegbusi-Slough, product manager, ARM, "When buying microcontroller software tools, a developer must consider the whole package offered by the tool, like compiler, simulator, support and ease-of-use." He adds, while selecting a compiler for embedded applications where memory sizes are restricted, developers look for a compiler that can generate the smallest, most efficient code.

An important part of developing an application is being able to test its behaviour, and optimise as required. Some developers are tempted by open-source tools, which are free to download and use. Such tools are not subject to regular maintenance, and have no dedicated support channels. Developers using such tools end up paying through down time experienced when they run into a bug. There is a high cost to time spent trying to get support from forums and chat rooms, or having to fix the tool themselves, or even resorting to switching to other tools partway through their project. Much time can be wasted at the start of a project trying to understand a new tool, and configure it for the device the application being developed is targeting.

Here in this article we discuss a few compilers and simulators that are available for programming of the microcontrollers.

Keil

The latest version of the Keil PK51 is v9.00. It supports the latest 8051 devices from vendors such as Evatronix, Nordic, NXP and SiLabs.

What's Special?

Keil development tools support the 8051 architecture and can be used by all, starting from school professionals to engineers. The Keil μ Vision Debugger simulates peripherals such as I²C, CAN, UART, SPI, interrupts, I/O ports, ADCs, DACs and PWM modules of your 8051 device.

Users' feedback

The software comes with a variety of 8051 microcontrollers. The code that the software understands is in 'C' language. Moreover, the software allows the developers to write code for Dallas 390 or Philips 51MX device.

Aiyegbusi-Slough explains that all Keil software tools benefit from regular updates to add support for the latest devices from their silicon partners, and example projects for supported devices. He adds that the Keil software development tools are used by a wide range of companies, from large silicon manufacturers such as Atmel, Infineon, NXP, STMicroelectronics and Texas Instruments to OEMs and end-users. The tools are used by many of the companies whose devices are listed in the Keil device database at 'http://www.keil.com/dd/'

Suggestion

The 8051 compiler, assembler, linker and debugger are limited to 2 kB of

object code. Moreover, the debugger supports programs that are 2 kB in size or smaller. Linker for code banking, library manager and RTX51 tiny real-time operating system are not included in the evaluation version. It would be great if the evaluation version could have the same features as the commercial one, even if these were available only for a limited number of days.

Home Page: <http://www.keil.com/>

Download URL: <https://www.keil.com/demo/eval/c51.htm>

Platform Supported: Windows

Pic Development Studio

Pic Development Studio is an open source software licensed under GPL. The latest version of the software is Pic Development Studio 1.1. The software is written in Java and runs on multiple platforms.

What's Special?

The Pic Development Studio basically works on Microchip PIC16F84 microcontroller. This software not only gives you the option of simulation but also a list of many peripheral components. The software includes input devices, display devices, logic gates, etc. Input devices have pushbutton, toggle buttons, menu keyboard and matrix button pad. The display devices contain 16x2 LCD, LED array, 7-segment display and 4x7 segment display. In the logic gates section, the software offers certain basic gates and 4511 BCD-to-7-segment decoder. The software also has an oscilloscope and a square tone generator.

Users' feedback

The software is easy-to-use and it has a short and powerful pictorial tutorial that explains things in a systematic manner. You can simulate your soft-